Approach Method for Entity-Aware Translation

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1 Task Definition and Objectives

The task to be tackled is to translate a given input sentence from the source language, which is English, to the target language, which we have chosen to be Chinese.

Named entities may be entities that are rare, ambiguous, or unknown to the machine translation system, so our task is to develop a machine translation system that can accurately translate such named entities in the input sentence to the target language.

1.1 Expected Difficulties

Ambiguity in Context: Entities often rely on context for correct interpretation. For instance, "Apple" could mean a fruit or the tech company, depending on the context. Ensuring the translation engine can discern which interpretation is correct is challenging, especially if the surrounding text provides minimal clues.

Complex Multi-word Entities: Names of companies, products, and events are often multi-word entities that can be difficult to translate correctly. Literal translations of parts of these names can lead to awkward or incorrect results. For example, "Department of Justice" should not be translated word-for-word but rather with an equivalent term that reflects the entity accurately in the target language.

Cross-lingual Consistency: Translating entities consistently across languages can be tricky, especially when a particular entity has multiple valid representations in the target language. For instance, translating the name of a brand or a product might require the MT system to stick with a specific convention, especially if that brand is globally recognized.

2 Dataset and Data Management

The dataset we are using provides a structured approach to multilingual translation and entity analysis, designed for tasks involving English source text translations to Chinese. Each data instance includes several key components, making it suitable for nuanced analysis of translation consistency and entity recognition across languages. The SemEval haven't released the final dataset yet, but there's sample dataset for reference.

Each entry in the dataset contains: **ID**: A unique identifier to track individual instances.
Source Locale: Always set to English (en) for this dataset, indicating that the source text is in English.
Target Locale: Specifies the locale for the target translations (e.g., Chinese, zh).
Source Text: A single English string serving as the original content.
Target Translations: A list of one or more possible translations in the target language. For example, for Chinese, multiple translations may be provided to capture variations in phrasing or nuance.

Mentions: A list of entities, such as named entities, identified in each target translation. (Currently, the sample dataset lacks this component, though it is expected in the final release).

Metadata: Includes supplementary details, such as a wikidata_id, to offer context about the source content.

For instance, a sample entry in the dataset for the Chinese target locale includes translations of the English question, "What are the consequences of the executive orders issued in Executive Orders?" into four possible Chinese variations: "《小说总统命令》中发出的命令结果是什么？", "总统命令释出的行政命令的后果是什么？", "总统命令中发布的行政命令的后果是什么？", "总统命令中释出行政命令的后果是什么？". The translations offer different phrasings to accommodate linguistic nuances, and metadata indicates the source context by referencing an external knowledge base (wikidata_id).

Given the partial release of the SemEval dataset, which lacks the complete set of entity mentions, the International Conference on Spoken Language Translation (IWSLT) 2017 dataset serves as a backup. The IWSLT dataset similarly involves English-to-Chinese translations and provides an alternative source of real-world conversational examples. "Thank you so much, Chris. And it's truly a great honor to have the opportunity to come to this stage twice; I'm extremely grateful.", "非常谢谢，克里斯。的确非常荣幸能有第二次站在这个台上的机会，我真是非常感激。"

3 Evaluation Metrics

3.1 METEOR

We decided to use METEOR as our evaluation metrics. METEOR is an automatic evaluation metric for machine translation that uses unigram matching to compare machine translations with human references. It matches unigrams by surface form, stemmed form, and meaning, allowing for additional advanced matching strategies. METEOR scores translations based on unigram precision, unigram recall, and fragmentation, the latter of which captures the word order accuracy in relation to the reference. Once all generalized unigram matches between the two strings have been found, METEOR computes a score for this matching using a combination of unigram-precision, unigram-recall, and a measure of fragmentation that is designed to directly capture how well-ordered the matched words in the machine translation are in relation to the reference.

3.2 BLUE

BLUE will be used to contrast. BLEU (Bilingual Evaluation Understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another. Quality is considered to be the correspondence between a machine' s output and that of a human: "the closer a machine translation is to a professional human translation, the better it is"–this is the central idea behind BLEU. BLEU was one of the first metrics to claim a high correlation with human judgements of quality, and remains one of the most popular automated and inexpensive metrics. Scores are calculated for individual translated segments—generally sentences—by comparing them with a set of good quality reference translations. Those scores are then averaged over the whole corpus to reach an estimate of the translation' s overall quality. Nei-

ther intelligibility nor grammatical correctness are not taken into account.

4 Proposed Approach and Methodology

The process of integrating entity-aware translation is divided into several steps to manage the complexities of accurately translating named entities from English to Chinese.

4.1 Separate Dataset Columns

The dataset, based on the IWSLT format, consists of English and Chinese sentence pairs. Examples include:

> {"Thank you so much, Chris. And it's truly a great honor to have the opportunity to come to this stage twice; I'm extremely grateful.", " 非常谢谢，克里斯。的确非常荣幸能有第二次站在这个台上的机会，我真是非常感激。"}

> {"I have been blown away by this conference, and I want to thank all of you for the many nice comments about what I had to say the other night.", " 这个会议真是让我感到惊叹不已，我还要谢谢你们留下的关于我上次演讲的精彩评论"}

To facilitate independent processing and entity matching across languages, the dataset is first organized into separate English and Chinese columns.

4.2 Named Entity Recognition (NER)

We utilize the Stanford Named Entity Recognizer (NER), a Java-based tool, to detect entities within both English and Chinese text. Specific pre-trained models are employed:

- **English Model:** `english.all.4class.distsim.crf`
- **Chinese Model:** `chinese.misc.distsim.crf`

Each word is tagged with one of four classes:

- **PERSON:** Names of individuals.
- **ORGANIZATION:** Names of organizations, such as companies or institutions.
- **LOCATION:** Geographic locations, including countries and cities.
- **GPE:** Political entities, encompassing countries, states, and cities.

Additionally, **two special tags** are used:

- **O:** Non-named entities.
- **MISC:** Miscellaneous entities that do not fit into specific categories.

	English Sentence	English Tags	Chinese Sentence	Chinese Tags
0	In Russia, all war monuments have this beautiful inscription at the top.	[(In, O), (Russia, LOCATION), (, O), (all, O), (war, O), (monuments, O), (have, O), (this, O), (beautiful, O), (inscription, O), (at, O), (the, O), (top, O), (., O)]	在俄罗斯，所有的战争纪念碑的顶部	[(在, O), (俄, GPE), (罗, GPE), (斯, GPE), (，, O), (所, O), (有, O), (的, O), (战, O), (争, O), (纪, O), (念, O), (碑, O), (的, O), (顶, O), (部, O)]
1	So, for example, I was giving this talk recently in New York City, and I said, "You know when you're on the subway and the other person across the subway car smiles at you, and you just instinctively smile back?"	[(So, O), (, O), (for, O), (example, O), (, O), (I, O), (was, O), (giving, O), (this, O), (talk, O), (recently, O), (in, O), (New, LOCATION), (York, LOCATION), (City, LOCATION), (, O), (and, O), (I, O), (said, O), (, O), (", O), (You, O), (know, O), (when, O), (you, O), ('re, O), (on, O), (the, O), (subway, O), (and, O), (the, O), (other, O), (person, O), (across, O), (the, O), (subway, O), (car, O), (smiles, O), (at, O), (you, O), (, O), (and, O), (you, O), (just, O), (instinctively, O), (smile, O), (back, O), (?, O), (", O)]	打个比方来说，我最近在纽约市演讲，其中说到："当你在地铁上，车厢对面的人 向你微笑时，你会下意识地回报以微笑？"	[(打, O), (个, O), (比, O), (方, O), (来, O), (说, O), (，, O), (我, O), (最, O), (近, O), (在, O), (纽, GPE), (约, GPE), (市, GPE), (演, O), (讲, O), (，, O), (其, O), (中, O), (说, O), (到, O), (：, O), (", O), (当, O), (你, O), (在, O), (地, O), (铁, O), (上, O), (，, O), (车, O), (厢, O), (对, O), (面, O), (的, O), (人, O), (向, O), (你, O), (微, O), (笑, O), (时, O), (，, O), (你, O), (会, O), (下, O), (意, O), (识, O), (地, O), (回, O), (报, O), (以, O), (微, O), (笑, O), (？, O), (", O)]

图 1: After Stanford NER

4.3 Entity Replacement with Temporary Tokens

Detected entities are replaced with temporary tokens in the format `<\tag class, text>` as shown in Figure 2. For instance, the word "Russia" is tagged as `<\LOCATION, Russia>` in English, and 俄羅斯 is tagged as `<\GPE, 俄 > <\GPE, 羅 > <\GPE, 斯 >`. These temporary placeholders enable sentence translation while

preserving entity placeholders, ensuring their integrity during translation.

	English Sentence	Tagged English Sentence	Chinese Sentence	Tagged Chinese Sentence
0	In Russia, all war monuments have this beautiful inscription at the top.	In <\LOCATION, Russia> , all war monuments have this beautiful inscription at the top .	在俄罗斯，所有的战争纪念碑的顶部	在 <\GPE, 俄> <\GPE, 罗> <\GPE, 斯> ，所有的战争纪念碑的顶部
1	So, for example, I was giving this talk recently in New York City, and I said, "You know when you're on the subway and the other person across the subway car smiles at you, and you just instinctively smile back?"	So , for example , I was giving this talk recently in <\LOCATION, New> <\LOCATION, York> <\LOCATION, City> , and I said , `` You know when you 're on the subway and the other person across the subway car smiles at you , and you just instinctively smile back ? ''	打个比方来说，我最近在纽约市演讲，其中说到："当你在地铁上，车厢对面的人向你微笑时，你会下意识地回报以微笑。"	打个比方来说，我最近在 <\GPE, 纽> <\GPE, 约> <\GPE, 市> 演讲，其中说到："当你在地铁上，车厢对面的人向你微笑时，你会下意识地回报以微笑。"

图 2: Put the token Back to the sentence

4.4 Model Definition for Translation

Two translation models are defined to handle distinct translation tasks:

- **Sentence Translation Model:** Handles translation of the main sentence structure.
- **Entity Translation Model:** Processes entities independently to maintain semantic accuracy.

Using separate models allows for tailored handling of sentence structure and context, enhancing translation accuracy.

4.5 Reconstruction of Translated Sentence

The translated sentence and entity tokens are merged, reconstructing a coherent translation with entities placed in their appropriate positions.

4.6 Stanford NER Model Insights

The Stanford NER model is pivotal in this approach as it provides accurate entity tagging. Each word is tagged separately, for example:

- "Russia" is tagged as a `LOCATION`.
- The special tag `O` denotes non-named entities, while `MISC` classifies untagged or miscellaneous entities.

4.7 Outcome and Current Status

The pre-processing tasks and initial model configurations have been completed, establishing a foundation for the remaining steps. The next phases will involve refining translation models and ensuring cohesive integration of tagged entities.

4.8 Evaluation

Our core evaluation metric is METEOR, and our contrasting metric is BLEU. By matching unigrams according to surface form, stemmed form, and meaning, METEOR evaluates the quality of machine translation. It quantifies word order accuracy in relation to the reference by combining fragmentation, unigram precision, and unigram recall. Because of this, METEOR is able to offer a complex score that takes into account both accuracy and fluency. In contrast, BLEU compares n-gram matches between machine and human translations to assess translation quality. It does not take grammatical accuracy or intelligibility into consideration, instead averaging segment-level ratings across the corpus.

5 Timeline

- **Project Proposal Report:** Nov 4
- **Completion of Initial Tasks:** Nov 11
- **Model Definition:** Nov 18
- **Final Task Completion:** Nov 25
- **Project Presentation:** Dec 2
- **Final Report Submission:** Dec 6

6 References